

Data Scientist (DS)

Module - 2

Module-End Assignment

Assignment Task Overview

Problem Statement

In this project, you will analyze an **online retail dataset** containing customer transactions, product details, quantities, prices, and purchase dates.

Businesses use such data to understand **sales trends, customer behavior, and product performance**. However, raw data requires cleaning, transformation, and analysis to generate meaningful insights.

You will build a **data analysis pipeline using NumPy, Pandas, Matplotlib, and Seaborn** to clean data, perform analysis, visualize patterns, and derive actionable business insights.

Objective

- Clean and preprocess real-world data
- Perform exploratory data analysis (EDA)
- Create new features for analysis
- Visualize trends using charts
- Generate business insights

Dataset: [Online Retail Dataset – UCI Repository](#)

Assignment Tasks:

Task 1 – Data Import & Setup

- Load dataset using Pandas
- Explore structure: `head()`, `tail()`, `shape`, `columns`
- Convert `InvoiceDate` to datetime

Task 2 – Data Cleaning

- Handle missing values (remove null `CustomerID`)
- Remove duplicates
- Fix invalid values (negative quantity, invalid price)

Task 3 – Feature Engineering

- Create $\text{TotalPrice} = \text{Quantity} \times \text{UnitPrice}$
- Extract time features (Year, Month, Day, Hour)
- Create categories (Customer Segment, Order Size, Day Type)

Task 4 – Data Exploration

- Use `describe()` and dataset overview
- Analyze categories (`value_counts()`, `unique()`)
- Perform `groupby()` (country, month, product)

Task 5 – Data Wrangling

- Aggregate data using `groupby()`
- Sort to find the top customers and countries
- Restructure data if needed

Task 6 – Statistical Analysis

- Analyze Quantity, UnitPrice, TotalPrice
- Calculate mean, median, and mode
- Find standard deviation, variance, and percentiles

Task 7 – Data Visualization (Min 8 Plots)

Matplotlib:

- Line chart
- Bar chart
- Histogram
- Box plot

Seaborn:

- Count plot
- Violin plot
- Heatmap
- Pair plot

Task 8 – Business Insights

- **Identify:**
 - Top country
 - Best sales month
 - Peak sales time
- **Analyze:**
 - Customer behavior
 - High-value customers
 - Top products

Task 9 – GitHub Project Implementation

- Create project folder structure:
 - `data/`, `notebooks/` or `scripts/`, `README.md`, `requirements.txt`
- Use VS Code for development
- **Initialize Git:**

```
git init
```

```
git add .
```

```
git commit -m "Initial commit"
```

- **Push to GitHub:**

```
git remote add origin <repo_link>
```

```
git push -u origin main
```

Maintain version control with proper commits

Deliverables:

Submit **GitHub Repository Link** containing:

- Complete Python code
- Cleaned dataset (if applicable)
- Visualizations (plots/screenshots)
- README file including:
 - Project description
 - Steps performed
 - Tools used
 - key insights

Skills to be Evaluated and Mapping

Assignment Section	Core Skill	Sub-Skill	Proficiency
Data Loading & Cleaning	Data Handling	Loading dataset, handling missing values, removing duplicates	Beginner–Intermediate
Feature Engineering	Data Transformation	Creating new columns and time-based features	Intermediate
Data Exploration & Wrangling	Data Analysis	Groupby, aggregation, sorting, restructuring	Intermediate
Statistical Analysis	Data Analysis	Mean, median, variance, distribution analysis	Intermediate
Data Visualization	Visualization	Matplotlib & Seaborn plots	Beginner–Intermediate
Business Insights	Interpretation	Identifying trends and patterns	Intermediate
GitHub Implementation	Version Control	Git commands, repo management, project structure	Beginner

Rubric & Mark Allocation (Total 25 Marks)

Skill / Component	Full Marks	Criteria for Full Marks	Partial Credit Description	Partial Marks Range
Data Loading & Cleaning	4	Dataset loaded correctly, missing values handled, duplicates removed, invalid data fixed	Minor issues in cleaning or validation	1 – 3
Feature Engineering	3	All required features (TotalPrice, time features, categories) created correctly	Some features missing or incorrect	1 – 2
Data Exploration & Wrangling	5	Proper use of describe(), value_counts(), groupby(), sorting, and merging	Partial analysis or incorrect aggregation	2 – 4
Statistical Analysis	3	Accurate calculation of mean, median, variance, and interpretation	Minor calculation or interpretation errors	1 – 2
Data Visualization	5	Minimum 8 plots created correctly with labels and clarity	Some plots missing or poorly labeled	2 – 4
Business Insights	3	Clear, meaningful insights derived from analysis	Basic or limited insights	1 – 2
GitHub Implementation	2	Proper repo setup, commits, structure, and README	Incomplete Git usage or poor structure	1

Skill Taxonomy Structure

- **Data Cleaning:** Handle missing values, duplicates, and invalid data
- **Feature Engineering:** Create derived columns and time features
- **Data Analysis:** Use groupby, aggregation, and statistical measures
- **Visualization:** Generate plots using Matplotlib and Seaborn
- **Interpretation:** Derive insights from data patterns
- **Version Control:** Manage project using Git and GitHub

Evaluation Guidelines

- Review the submitted **GitHub repository/notebook**
- Ensure:
 - The dataset is correctly loaded and cleaned
 - Features are created properly
 - Analysis and visualizations are complete
 - Code runs without errors
- **For each section:** Map work to defined skills and assign marks based on the rubric
- **Marking Criteria:**
 - Full marks: Correct logic, clean code, proper visualization
 - Partial marks: Minor errors or incomplete work
- **Additional Checks:**
 - Plots are clear and labeled
 - Code is readable and structured
 - The GitHub repo is organized with commits

Achievement Criteria & Passing Rules

- Minimum marks required in all sections
- Must demonstrate at least two key skills:
 - Data Cleaning
 - Data Visualization
- Project must be:
 - Executable
 - Properly structured in GitHub
- **Feedback:** Provide a **detailed scorecard with strengths and improvements**